

Numerical Modeling and Experimental Measurements of the Electric Potential Generated by Cochlear Implants in Physiological Tissues

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Abstract—The electric potential pattern generated by a cochlear implant was modeled using the finite-element method. The pattern, which is a response to bipolar current pulses from the implant electrode array, was investigated for different values of the electrical parameters of the human cochlea and of the array. Simulated data are compared with measurements in a water tank. Numerical simulations were carried out for different electrode configurations. Electrode configurations were found to affect the resulting potential, and the measurements confirmed our modeling approach. This study provides a better understanding of the relationship between the stimulation parameters and the delivered electric field, which is crucial to develop more efficient and spatially localized excitations of cochlear neural tissues.

Index Terms—Biomedical measurements, finite element methods, hearing aids, implantable biomedical devices, voltage measurement.

I. INTRODUCTION

A COCHLEAR implant is a prosthetic device implanted into the inner ear, i.e., the cochlea. It is used to directly stimulate the neural tissue of the spiral ganglion (which is the inferior root of the acoustic nerve) with electrical pulses. The neural discharges resulting from such electrical stimuli induce auditory sensations in a brain cortex area and can restore partial hearing to deaf people with severe to profound deafness [1].

The electrodes implanted into the cochlea stimulate the ear nervous terminals by means of a series of bipolar current pulses, whose amplitude, width, and frequency are controlled by a speech processor. To maintain consistency with the standard terminology used in clinical practice and in the literature on cochlear implants, we will use throughout the paper the term “biphasic pulses” instead of “bipolar pulses.”

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In multichannel implants (i.e., those with multiple electrodes), the electrode array excites simultaneously auditory nerve fibers at different locations inside the inner ear, thus mimicking coding acoustic frequencies. Which electrodes are activated depends on the frequency components contained in the spectrum of the incoming speech signal [2].

It is well known that stimulation parameters, such as the configuration of the electrode array, stimulus intensity, pulse stimulation rate, and pulse duration, affect, through the generated electric field, the thresholds for electrical neural stimulation, the number of fibers that are excited or hyperpolarized, the initiation site of excitation, and the patient pitch perception [3]. A deeper knowledge of the relationship between the stimulation parameters and the electric field in the physiological tissue is crucial to develop more efficient and spatially focused excitations of cochlear neural tissues.

The aim of this study is a systematic investigation, both numerical and experimental, of the relationship between the electric fields generated in cochlear tissues around the electrode array and the stimulation parameters. The distribution of the electric potential in the cochlea was simulated by 3-D mathematical modeling the electrical properties of the human cochlea and of the electrode array. Simulated data are compared to the measured potential distributions in a water tank, in the 3-D space around the stimulating electrodes.

II. PROPOSED APPROACH

Numerical computation of the electric potential—The 3-D electric potential induced by the cochlear implant stimulation inside the inner ear was calculated by means of the finite-element method (FEM), with the help of a commercial software (Maxwell 3-D v6.0, Ansoft, Pittsburgh, PA). The electric potential Φ generated by the cochlear implant stimulation inside conduction volume (i.e., the cochlear structures) is given by [4]

$$\Phi(x', y', z') = \frac{1}{4\pi(\sigma + j\omega\epsilon)} \int_V \frac{I_V(x, y, z)e^{-jkR}}{R} dV \quad (1)$$

where I_V is the current source per unit volume provided by the cochlear implant stimulation, σ and ϵ are the conductivity and dielectric permittivity of the medium, ω is the angular frequency of the excitation current, (x, y, z) and (x', y', z') are the 3-D coordinates of the current source and of the point where the potential is computed, $R^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$, $k^2 = -j\omega\mu\sigma_c$, where μ is the permeability of the medium,

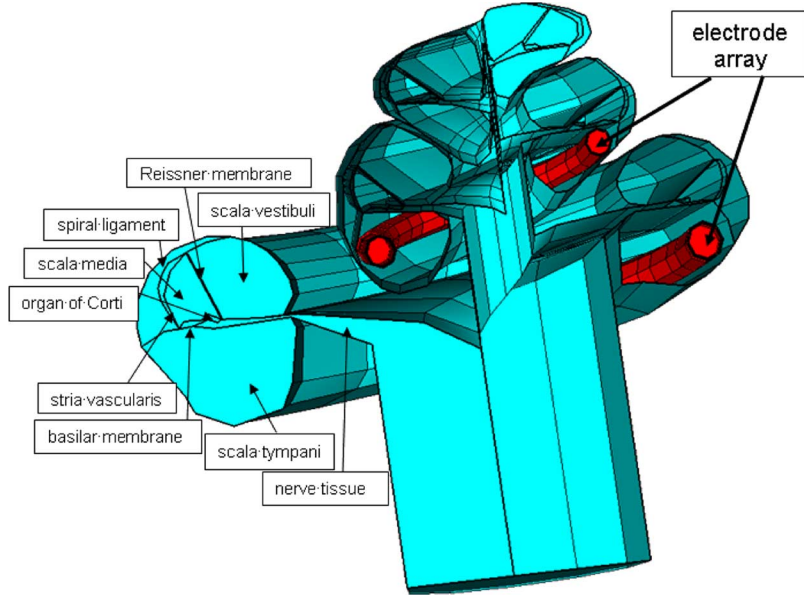


Fig. 1. Sagittal section of the electrode array and of the 3-D realistic model of a human cochlea (volume 84 mm³, 130 000 tetrahedra) obtained from the extrusion around the longitudinal axis of a midmodiolar section of a real cochlea.

and σ_c is a complex conductivity that includes pure conductive effects, dielectric losses, and dielectric displacement, defined as

$$\sigma_c = \sigma \left(1 + \frac{j\omega\epsilon}{\sigma} \right). \quad (2)$$

To simplify (1), a quasi-static approximation can be introduced, which assumes that at each instant of time the electric potential field satisfies Poisson's equation, and that the boundary conditions correspond to a stationary condition. This quasi-static approximation can be applied if [4]:

$$\begin{aligned} \omega\epsilon/\sigma &\ll 1 \text{ (i.e., capacitance effects can be neglected);} \\ kR_{\max} &\ll 1 \text{ (i.e., propagation effects can be neglected);} \\ (kR_{\max})^2 &\ll 1 \text{ (i.e., inductive effects can be neglected).} \end{aligned}$$

These assumptions are valid in the frequency range associated to a typical current stimulus of a cochlear implant device (from 20 to 80 kHz), for the dielectric properties of cochlear tissues [6]–[9], and for $R_{\max} = 0.004$ m (where $R_{\max}$ is the maximum value of R to be considered in the conductor volume V).

Under the quasi-static approximation, (1) becomes

$$\Phi(x', y', z') = \frac{1}{4\pi\sigma} \int_V \frac{I_V(x, y, z)}{R} dV \quad (3)$$

which is equivalent to the Poisson equation

$$\nabla^2 \Phi = -\frac{I_V}{\sigma}. \quad (4)$$

It is worth noting here that while the quasi-static assumption is valid in the 20–80-kHz range, it can not describe the evolution of higher frequencies components of the electric stimulus due to the fast pulse edges. This will be the object of further studies.

In practice, the FEM approach is used to solve (4) by subdividing the conduction volume V into small tetrahedra. The potential Φ is calculated at the vertices of each tetrahedron and interpolated elsewhere. The solution process involves a progressive refinement of the mesh subdivision of the volume V until

a percent error energy, which is a measure of how closely the solution satisfies (4), is less than the specified target value. In this study, numerical simulation were computed using an energy error $\leq 2\%$.

A. Cochlear and Electrode Array 3-D Models

A 3-D synthetic geometric model resembling the true shape of human cochlea structures was developed. The model is obtained by the 3-D extrusion around the longitudinal axis of the cochlea—the modiolus (i.e., the bony conical central axis of the cochlea) of a 2-D cross section of a human cochlea, from [10]. The extrusion is obtained by rotating and replicating the 2-D contour of the cochlea every 30° and by connecting with straight lines the points of each contour. The model thus obtained is shown in Fig. 1 and is made of 130 000 tetrahedra. The cochlear model of Fig. 1 consists of eight different structures that characterize a real human cochlea: scala tympani and vestibuli (both filled with perilymph); scala media (filled with endolymph); spiral ganglion; peripheral axonal processes; Reissner's membrane; basilar membrane; stria vascularis; and organ of Corti. Conductivity of materials of the cochlear structures were based from [6]–[9]. Finally, the cochlear model was embedded into a bone cylinder (conductivity $\sigma = 0.156 (\Omega \times \text{m})^{-1}$) with a radius of 20 mm and height of 20 mm.

The electrode array was modeled according to the size, shape, and materials of the true Nucleus CI24M cochlear implant array [11], which consisted of 22 equally spaced platinum electrode bands and additional stiffening rings on a flexible silicone carrier. Each electrode is connected independently to the current source generator by individual and insulated platinum-iridium wires. The length of the whole electrode array is 17 mm; each ring electrode has a diameter of 0.4–0.6 mm and height of ~ 0.4 mm. Conductivities of the platinum electrodes were $10^6 (\Omega \times \text{m})^{-1}$ and of the silicone carrier $10^{-15} (\Omega \times \text{m})^{-1}$.

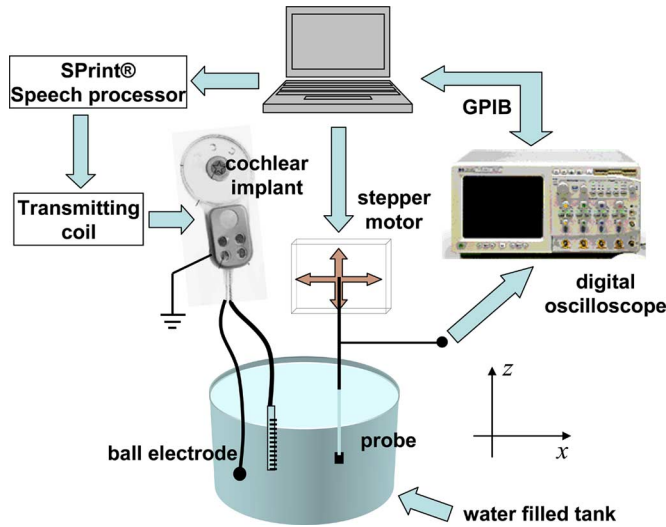


Fig. 2. Experimental setup for PC-controlled potential measurement.

The position of the electrode array inside the scala tympani was chosen according to the mean insertion path data derived from Computed Tomography scans of the petrous bone (i.e., that portion of the temporal bone where the cochlea is located) of patients implanted with a cochlear implant [12].

Two different electrode configurations used in clinical practice were considered: 1) the monopolar (MP) mode, where a single electrode inside the cochlea is active and the electrical ground (“indifferent electrode” in medical language) is located outside the cochlea in a depression of the mastoid bone and 2) the bipolar (BP) mode, where two electrodes in the array are active with the same intensity and opposite polarity (in this case the positive one is “active” and the negative one is named “indifferent”). In each configuration the ground reference of the implant is connected to the “ball electrode” (Fig. 2), an electrode placed in mastoid bone by the surgeon during the surgical operation.

B. Experimental Setup

Electric potentials produced in a $5.0 \times 7.5 \times 3.5 \text{ cm}^3$ water-filled tank by the electrode array of the cochlear implant are measured. A cochlear implant PC programming platform (WinDps by Cochlear, Lane Cove, NSW, Australia) was used to drive the electrode array and generate charge balanced biphasic pulses with $100 \mu\text{s}$ of pulse width, $25 \mu\text{s}$ of phase gap (which is the time interval between the negative and positive phase of the biphasic pulse), and $120 \mu\text{A}$ of current. Bipolar electrode configuration was tested with electrode #18 “active” and electrode #21 “indifferent.”

Fig. 2 shows the experimental setup. The cochlear implant was placed above the tank and the electrode array was placed in the water tank vertically. The voltage produced by the electrodes in the water tank was recorded by an insulated probe (a copper wire of 0.3-mm diameter) whose position was controlled by an automated 3-D stepper motor (8MT173-20-50 by Standa, Vilnius, Lithuania), with a step resolution of $1.25 \mu\text{m}$. The difference potential measured by the probe was detected with a 100-MHz digital oscilloscope (TDS210 Tektronix, Beaverton, OR) and stored with 8 bits per sample on a PC. The above band-

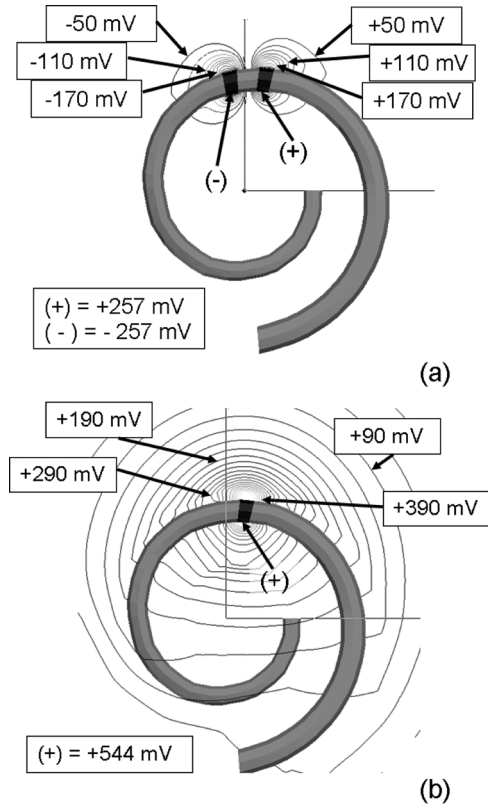


Fig. 3. Potential distribution (equipotential contours) on an horizontal section of the cochlea through the active electrode for 1-mA BP stimulation (a) and for 1-mA MP stimulation (b). In both panels, equipotential contours are spaced by 20 mV and the active electrode is #10. In the BP configuration, the “indifferent” electrode is #11. The spiral-shaped gray object is the electrode array. The positions of the maximum and minimum of the potential are denoted by the symbols (+) and (-). In each panel, the legend shows the value in mV of the maximum and minimum potentials.

width was needed to observe the rise and fall time of the biphasic stimuli.

The ground reference of the probe was in the water tank a few centimeter from the array. The potential was measured at 0.5-mm steps along the vertical z - and horizontal x -axes. The 3-D voltage distribution was deduced from the above using cylindrical symmetry of the simplified system. In order to increase the measurement accuracy, an averaging over 64 samples was implemented. At each position of the stepper motor, the system recorded the biphasic potential waveform from the implant.

The measurement was completely automated using a custom-made LabVIEW Virtual Instrument, which controlled the stepper motor and, the digital oscilloscope. It recorded the position of the probe and the electric potential waveform.

III. RESULTS

Figs. 3 and 4 show the equipotential contours calculated inside cochlea when delivering a stimulus of 1 mA with the BP [see Fig. 3(a) and 4(a)] and MP [Fig. 3(b) and 4(b)] configurations. In these examples, the potential was calculated by subdividing the conducting volume of the cochlea ($13 \times 13 \times 9.7 \text{ mm}^3$) in $\sim 130\,000$ tetrahedra, 7100 of which were used to model the electrode array.

The electrode configuration (BP or MP) deeply influences both the amplitude and spatial distribution. As it is possible to

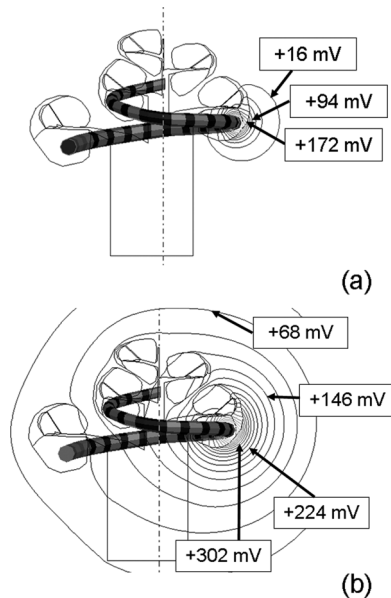


Fig. 4. Potential distribution (equipotential contours) on a sagittal section of the cochlea passing through the active electrode for 1-mA BP stimulation (a) and for 1 mA MP stimulation (b). In both panels, equipotential contours are spaced by 26 mV and the active electrode was #10. In the BP configuration the ‘indifferent’ electrode is #11. The spiral-shaped gray object is the electrode array. The maximum and the minimum of the potential were the same as in previous Fig. 3.

see in Figs. 3 and 4, the MP configuration [Figs. 3(b) and 4(b)] generate higher values of the electric potential with respect to the BP mode [Figs. 3(a) and 4(a)]. Namely, with the MP configuration, the maximum of the electric potential is 544 mV, whereas with the BP configuration the maximum of the potential is lower and equal to 257 mV. This is due to the impedance seen by the current that flows from the electrode. With the BP configuration, the impedance seen by the current is the one in the volume between “active” and “indifferent” electrodes. In MP configuration, the impedance seen by current is the one from the “active” electrode and the “ball” electrode. This causes the different amplitude of the potential in the two different configurations.

Differences are observed also in the shape of the equipotential contours. The BP mode produced the more localized stimulation. The potential reaches the maximum at the active electrode and falls off rapidly with the distance from it. Conversely, the MP mode produces less localized pattern.

Differences in equipotential contours are even more evident when the electric potential is computed for a spiral-shaped path, as shown in Fig. 5. The potential was calculated along a spiral path close to the array on the inner wall of the scala tympani, just beneath the basilar membrane and close to the electrode array and the nerve. The path corresponds to the location of the first synapses of the afferent fibers of the auditory nerve. It is seen that the highest potential is generated by the MP configuration, whereas the BP configuration produces a bipolar potential (i.e., with a positive and negative peak) with lower amplitude. Along the path, the maximum amplitude of the BP voltage is about 1/9 of the MP response (29 mV of peak voltage compared to 257 mV) and the MP voltage is about half of the single electrode maximum (253 mV of peak voltage compared to 544 mV).

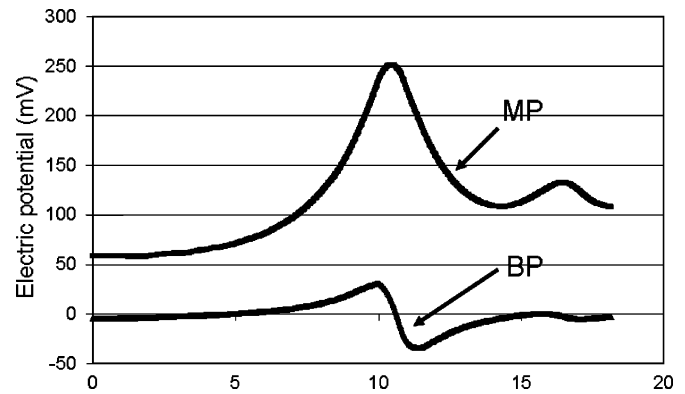


Fig. 5. Calculated distribution of electric potential on a spiral-shaped path along the basilar membrane for the BP and MP electrode configurations (as indicated by the arrows).

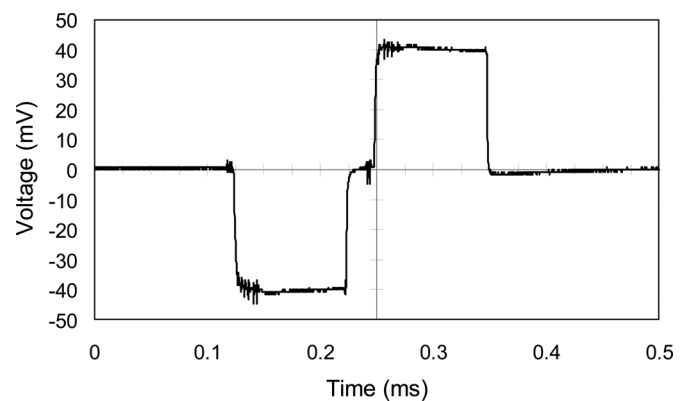


Fig. 6. Experimental biphasic potential waveform ($x = 500 \mu\text{m}$, $y = 0 \mu\text{m}$, $z = 1000 \mu\text{m}$). The current stimulus generating the measured potential was delivered using the BP electrode configuration with 100- μs pulse width, 25- μs phase gap, and 120- μA current.

Measurements of potential pattern around the electrode array in water bath were done both in the axial direction (along the line connecting all the electrodes in the array, z -axis) and in the transversal direction (x -axis). Results presented in this study were obtained for the BP electrode configuration, since with this stimulation it is easier to measure antisymmetric voltage around the central point between the two active electrodes. The standard deviation of the waveform amplitude was 1.9 mV based on ten trials with a measurement resolution of 0.4 mV. Fig. 6 shows, as an example, the waveform of potential measured in the water tank. In this example, the potential was measured at about 0.5 mm from the active electrode. A characteristic biphasic shape of the measured potential is seen with the negative phase leading.

The voltage amplitudes sampled at the various points of the 2-D grid in the $x - z$ plane are presented as a 3-D plot in Fig. 7. As expected, the maximum and minimum of the potential are found near the active and indifferent electrodes, respectively. The extrema rapidly decrease away from the electrode array along the transverse direction.

These results were used to perform a first validation of the potential distribution obtained with the numerical FEM model. Fig. 8 shows the peak-to-peak potential when measuring probe was moved along the z axis at a 500- μm distance from the active

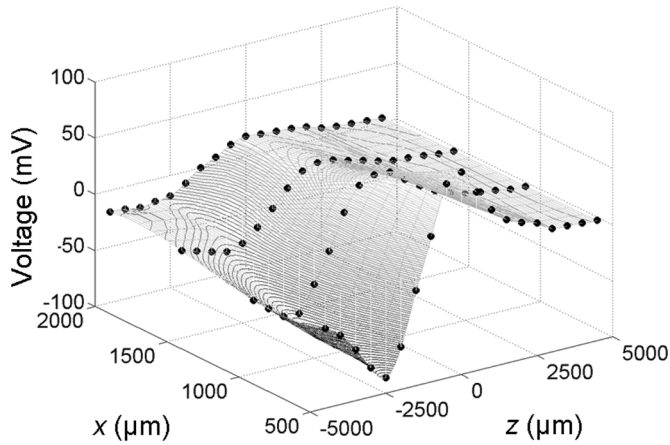


Fig. 7. 3-D plot of the electric potential. The sampling step in both the longitudinal and transversal direction is $500\ \mu\text{m}$. Experimental data (shown with the symbol $(\bullet)$) were used to fit a spline surface at $100\ \mu\text{m}$ sampling step with an interpolation polynomial of third order.

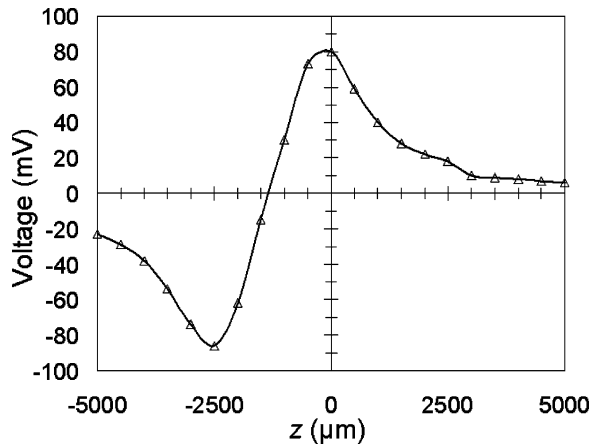


Fig. 8. Electric potential measured along the z -axis at a distance of $500\ \mu\text{m}$ from the active (positive) electrode. Triangles are the measured values.

electrode. It should be noted that the potential distribution measured through the FEM simulation along the spiral-shaped path near the nerve tissue (see Fig. 5) has a very similar shape to that measured experimentally and shown in Fig. 8. As expected, the simulated potential exhibits the same bipolar morphology and spread of the electric potential as the one effectively measured in the tank. In good agreement with theoretical modeling and numerical simulations, the potential experimentally measured exhibits a good symmetry and width between the negative and positive peaks. Some differences are found between the peak values of the potential measured experimentally and those simulated by FEM but these differences are due to the different position where potentials were recorded: the simulated potential was calculated at the interface between the perilymph ($\sigma_{\text{perilymph}} = 2\ \text{S/m}$) and the nervous tissue ($\sigma_{\text{nerve}} = 0.3\ \text{S/m}$); the measured potential was taken in drinking water contained in the tank ($\sigma_{\text{water}} = 0.05\text{--}0.005\ \text{S/m}$).

IV. CONCLUSION

A realistic 3-D model of the cochlear volume conduction was developed to simulate the potential distribution generated by

the electrical stimulation of a cochlear implant array. Numerical simulations were performed by means of the FEM approach and were computed using different configurations of stimulating electrode, namely, the MP and BP configurations. The model was extremely useful to quantitatively estimate the different focusing of the field produced by the two tested configurations. The BP mode produced the more localized stimulation. The potential reaches the maximum at the active electrode and falls off rapidly with the distance from it. Conversely, the MP mode produces a less localized pattern.

An experimental setup was also designed to measure the distribution of the potential released in water solution by the array. Peak-to-peak potential values were measured experimentally in the 3-D volume around the array using a probe moved with $500\text{--}\mu\text{m}$ resolution step. Our experimental measurement system was accurate ($<2\%$ voltage uncertainty and $<0.1\%$ time interval uncertainty) and could measure all the waveforms actually delivered at the implant electrodes.

Potential distributions calculated through the FEM approach were validated with the experimental data. For comparison purposes, experimental measures were performed with the same electrode array as that modeled in FEM simulations. Simulations derived from FEM modeling were highly accurate as revealed by the good agreement between experimental and simulated data. As such, the proposed model might be useful to quantitatively estimate the effects of configuration, position, material and shape of the array on the excitation field produced by the cochlear implant. These results might give an important contribution to a deeper understanding of the mechanism of the electrical auditory nerve stimulation and to improve the development of optimized auditory prostheses.

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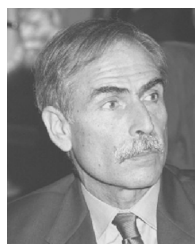
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